

SFNet-N: An Improved SFNet Algorithm for Semantic Segmentation of Low-Light Autonomous Driving Road Scenes

Hai Wang^{ID}, Senior Member, IEEE, Yanyan Chen, Yingfeng Cai^{ID}, Senior Member, IEEE, Long Chen^{ID}, Yicheng Li, Miguel Angel Sotelo^{ID}, Fellow, IEEE, and Zhixiong Li^{ID}, Senior Member, IEEE

Abstract—In recent years, considerable progress has been made in semantic segmentation of images with favorable environments. However, the environmental perception of autonomous driving under adverse weather conditions is still very challenging. In particular, the low visibility at nighttime greatly affects driving safety. In this paper, we aim to explore image segmentation in low-light scenarios, thereby expanding the application range of autonomous vehicles. The segmentation algorithms for road scenes based on deep learning are highly dependent on the volume of images with pixel-level annotations. Considering the scarcity of labeled large-scale nighttime data, we performed synthetic data collection and data style transfer using images acquired in daytime based on the autonomous driving simulation platform and generative adversarial network, respectively. In addition, we also proposed a novel nighttime segmentation framework (SFNET-N) to effectively recognize objects in dark environments, aiming at the boundary blurring caused by low semantic contrast in low-illumination images. Specifically, the framework comprises a light enhancement network which introduces semantic information for the first time and a segmentation network with strong feature extraction capability. Extensive experiments with Dark Zurich-test and Nighttime Driving-test datasets show the effectiveness of our method compared with existing state-of-the-art approaches, with 56.9% and 57.4% mIoU (mean of category-wise intersection-over-union) respectively. Finally, we also performed real-vehicle verification of the proposed models in road scenes of Zhenjiang city with poor

lighting. The datasets are available at <https://github.com/pupuchenyan/semantic-segmentation-on-nighttime>.

Index Terms—Semantic segmentation, deep learning, low visible light, autonomous driving.

I. INTRODUCTION

DUE to the rapid advancements in deep neural networks, semantic segmentation has made tremendous progress in various fields, such as autonomous landing of unmanned aerial vehicles, medical imaging, and autonomous driving. In particular, it has significantly enhanced the perception ability of unmanned vehicles. For the intelligent vehicle environmental perception, semantic segmentation unifies different detection tasks in an efficient way, thus avoiding the complexity induced by multi-sensor fusion. Image segmentation is essentially a fine unit-pixel regression task, which consists of classifying each pixel in an image, such as mapping the background to 0 and the foreground to one of the $N-1$ classes.

Currently, the convolutional neural network (CNN) based algorithms, such as Deeplab [1]–[4] or HRNet-OCR [5], [6], perform segmentation tasks with high accuracy and fast inference speed. However, most of these methods are based on images captured by cameras under good lighting and fair weather conditions. In unfavorable conditions, such as low lighting, rain, and fog, the segmentation performance degrades significantly, due to the decline in the overall image contrast and increased ambiguity of the object boundary. Moreover, it is impossible to avoid such conditions in real-world applications. In this work, we focus on solving the problem of semantic segmentation in low illumination scenes.

In low-light scenes, the images acquired using cameras may suffer from underexposure, noise, and motion blur. Notably, the features extracted by the convolution layers from such images and those acquired under good light conditions are quite different in terms of structure and texture. Therefore, the models trained using the authoritative daytime datasets, such as CityScapes [7], cannot be directly applied to low-light scenarios. Moreover, deep learning is essentially based on data-driven methods. The current standard strategy for achieving high-performance semantic segmentation is to train a massive amount of annotated low-light real-world images using neural networks. However, the process of collecting such low-light images and annotating them is labor intensive. In addition, due to the large area of dark or shaded parts

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Hai Wang and Yanyan Chen are with the School of Automotive and Traffic Engineering, Jiangsu University, Zhenjiang 212013, China (e-mail: wanghai1019@163.com; 1912128283@qq.com).

Yingfeng Cai, Long Chen, and Yicheng Li are with the Automotive Engineering Research Institute, Jiangsu University, Zhenjiang 212013, China (e-mail: caicaixiao0304@126.com; chenlong@ujs.edu.cn; liyucheng070@163.com).

Miguel Angel Sotelo is with the Department of Computer Engineering, University of Alcalá, 28801 Alcalá de Henares (Madrid), Spain (e-mail: miguel.sotelo@uah.es).

Zhixiong Li is with the Faculty of Mechanical Engineering, Opole University of Technology, 45-758 Opole, Poland, and also with the Yonsei Frontier Laboratory, Yonsei University, Seoul 03722, Republic of Korea (e-mail: zhixiong.li@yonsei.ac.kr).

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in night pictures, it is very difficult to establish high-quality annotations at pixel level. One solution to this problem is to train synthetic data instead. Therefore, we used the autopilot simulation platform to simulate low-light scenes in seven towns, under four weather conditions, and used the on-board visible light camera of the simulator to collect the corresponding synthetic data. It should be noted that there are still differences between the synthetic and actual scene data in terms of structure, color, and other features. Based on this characteristic, we also used generative adversarial networks (GANs) [8] to acquire style-transformed images using the existing well-lit public datasets, and obtained low-lit patterns which fully retained the characteristics of the real scene.

Due to insufficient illumination and artificial light sources, the nighttime images have some problems, such as information loss and small difference between classes. It is not appropriate to directly use the general segmentation network in daytime. Therefore, we proposed a nighttime segmentation framework (SFNET-N) for specific low-light conditions, which can effectively learn the hidden features in darkness. Our method mainly includes two networks: a light enhancement network and a segmentation network. The light enhancement network learns a set of optimal illumination fitting curves from pixel-to-pixel and semantic views, which can reconstruct details in darkness without distortion. At the same time, the designed segmentation network fully excavates the contextual information of pixels and minimizes the loss of information in the decoder. This can obtain dense feature representations and recover details as much as possible. Moreover, we also used the optimized multi-scale attention mechanism to further improve the performance. Additionally, we conducted extensive experiments using Dark Zurich-test [9] and Nighttime Driving-test [10] datasets to verify the effectiveness of our proposed model for nighttime semantic segmentation. The major contributions of this work are as follows:

- 1) Considering the scarcity of labeled nighttime data, two proprietary low-light automatic driving road scene datasets were constructed, namely Synthesiscarla and Cyclecity. These datasets were generated using the automatic driving simulation platform and GANs, respectively.
- 2) We proposed a unified segmentation framework for low illumination scenes, which includes a light enhancement network and a segmentation network.
- 3) The experimental results of Dark Zurich-test and Nighttime driving-test demonstrated that the proposed model outperformed the existing state-of-the-art methods on nighttime scenarios. Moreover, it also achieved excellent real-vehicle segmentation results for real-world scenes.

II. RELATED WORKS

A. Semantic Segmentation

Semantic segmentation is a challenging task in computer vision. Various methods have been reported in literature to perform semantic segmentation. Most of these methods are designed using CNN paradigm in deep learning and perform efficiently. For example, AlexNet [11] won the ImageNet

image recognition competition. Long *et al.* presented the fully convolutional neural network (FCN) [12], which is a pioneering work in the field of image segmentation. The FCN replaces the fully connected layers used for classification into fully convolutional layers, which allows the network to efficiently generate segmentation masks for a resolution of any size. The subsequent semantic segmentation models, such as UNet [13], UNet++ [14], DenseCRF [15], DLNet [16] and PSPNet [17] are based on FCN.

The core idea for improvement is devising a process for obtaining high pixel representations, while maintaining high image resolution. HRNet [5] maintained high-resolution representations in the entire network by connecting multiple resolution subnets in parallel, and then repeatedly integrating each subnet with other parallel subnets to obtain dense high-level features. SegNet [18] used the encoder-decoder to decode high-level feature maps to the original input size through up-sampling or deconvolution. Then, the loss of detailed information such as object boundary caused by simple operations was compensated through skip-connections. The context-based approaches [3], [17] have shown promising performance. For example, one study [3] presented the ASPP module, which used parallel dilated convolutions [19] with different dilation rates to aggregate multi-scale context information. Moreover, the receptive field was expanded without increasing the number of parameters to obtain denser high-level feature maps.

Several works have also been reported [20]–[25] on attention mechanism. For instance, DANet [22] effectively modeled the features from the spatial and channel perspectives to capture long-range dependencies. CCNet [24] proposed a criss-cross attention module to aggregate long-range dependencies from horizontal and vertical directions. Recently, there has been a huge upsurge [26]–[29] in studies applying Transformers, originally used for Natural Language Processing (NLP), to visual tasks. SETR [26] was the first to use Transformers to build an end-to-end segmentation framework, adopting ViT [29] as a backbone to extract features. SegFormer [28] proposed a simpler, more efficient framework that combines hierarchically structured Transformers with multi-layer Perceptron (MLP) decoders, and eliminated positional encoding in the encoder. Specifically, all the above improvements are based on good illumination and weather conditions by default. Instead, we intend to address image segmentation at nighttime in this work.

B. Semantic Segmentation in Low Light Scenes

Since universal CNNs can only learn knowledge from a specific domain, the model trained in the source domain usually has poor results in the target domain with inconsistent data distributions. This means that the daytime model cannot be directly applied to nighttime scenes. Recently, domain adaptation methods based on transfer learning have been used to narrow the domain gaps. The Dark Model Adaptation (DMA) [10] divides the twilight time between day and night into three subgroups based on the elevation of the sun, and makes a three-step incremental adjustment to the network according to the pseudo labels. This allows the knowledge to be transferred from day to night in an unsupervised manner.

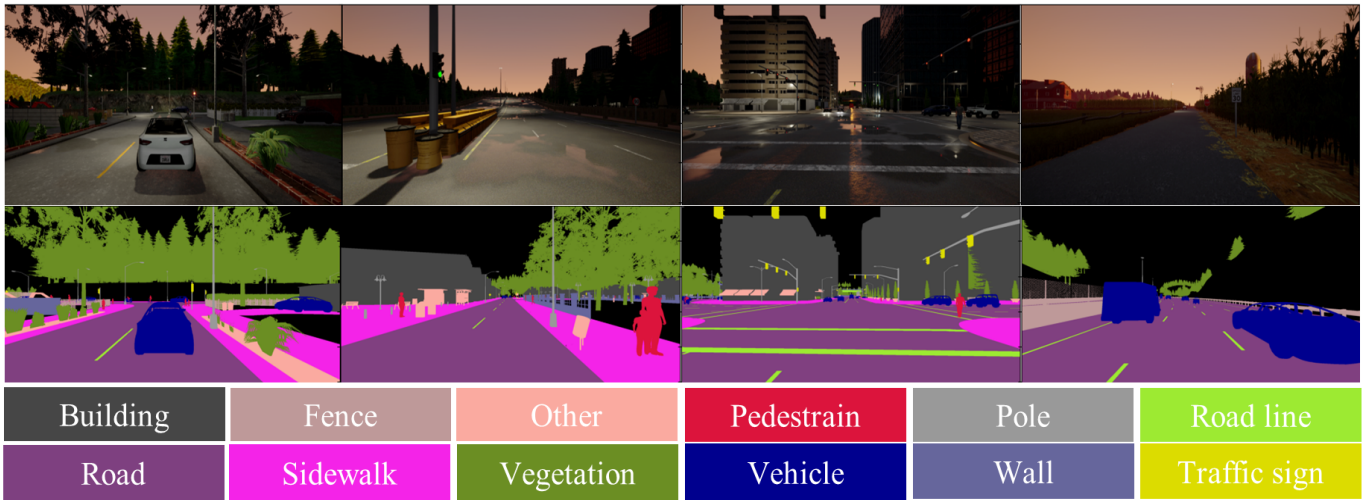


Fig. 1. Several samples for different road scenarios and weather conditions from our Synthesiscarla dataset.

CDAda [30] is a curriculum domain adaptation method, which adjusts the model by entropy minimization and pseudo-label self-training method respectively according to the difficulty of the domain. Another study [31] proposed an appearance transferring module to learn the shared feature representation of day-night images, thus improving the performance of night segmentation.

Some recent works also use GANs [8] to reduce the domain gaps effectively. For example, two separate studies [32] and [33] used GANs to learn the mapping of input images to output images, and improved the segmentation performance from two perspectives, including direct inference of night images and real-time online conversion inference of night images through style conversion. DANNet [34] proposed a one-stage domain adaptation network, which adopted an adversarial training with labeled daytime data and unlabeled roughly aligned day-night image pairs. For a variety of adverse conditions, a previous study [35] adapted the daytime model to rainy night scenarios by employing the idea of a pair of near scene cross-weather images sharing similar spatial layout and local context.

All the above methods are used with unsupervised learning or semi-supervision learning. A previous work [36] proposed a real nighttime dataset with dense annotations and an exposure aware framework to guide the network to explicitly learn exposure features. Unlike the above study [36], our proposed datasets are from the perspective of synthesis and migration, which greatly reduces human and material resources. Additionally, our proposed network is mainly composed of two parts: an image enhancement module and a segmentation module.

III. THE DATASETS FOR TRAINING

The datasets required for modeling different scenes cannot be universal. Therefore, different datasets and network architectures are required for different scenarios. It means that a large number of images with intensive pixel-level annotations must be collected to match the model. Existing semantic segmentation datasets are mainly collected on well-illuminated images, including some or few nighttime images.

It is noteworthy that most of the datasets with poor light used by researchers are self-collected through the surveillance videos captured by the transportation department and on-board cameras. Afterwards, a few data are annotated manually according to the research needs. In order to achieve better night segmentation performance, this work still follows the supervised learning paradigm.

Considering the annotation cost required to build an accurate and effective low-illumination dataset comprising real scenes, and the limitation of hardware resources in our laboratory, we used the simulation platform CARLA [37] to collect the low-light dataset required for this experiment and named it Synthesiscarla. In addition, its pixel-level annotations are completed by partially automating the process for reducing the cost of annotations. Meanwhile, in order to ensure the diversity of data, we collected data from seven different urban and rural scenarios using CARLA 0.9.9 software, including different types of urban streets, highways, tunnels, and narrow rural roads. For simulating realistic scenes more vividly, four weather conditions, including sunny, rainy, foggy, and cloudy, were also considered during the collection. The low-illumination data for different road scenarios and different weather conditions are shown in Fig.1. In order to conduct more comprehensive and fairer experiments and increase the diversity of datasets, we also used CycleGAN [38] algorithm for low-light processing of existing authoritative daytime datasets. The new dataset is called Cyclecity. In order to further improve the segmentation results, we also used the joint training strategy, training only 345 nighttime images in Bdd100K and 400 nighttime images in ACDC with the transferred results. The details of each dataset are as follows.

A. Synthesiscarla

The dataset is divided into training set and validation set, which contain 3338 images and 371 images with high-quality pixel-level annotations respectively, according to the dataset format of Cityscapes. Each image has a resolution of 1024×2048 pixels. More than 3K images are annotated with high-quality pixel-level annotations according to the dataset format of Cityscapes. The low-light scenes are roughly divided

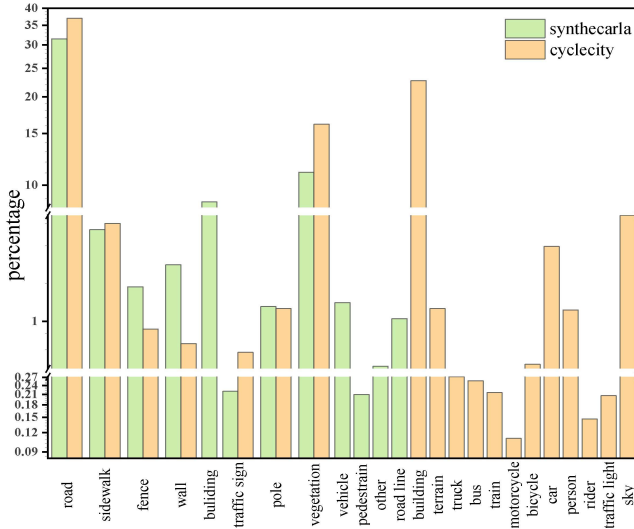


Fig. 2. The pixel distributions of our two datasets. The ordinate represents the proportion of pixels of each class in total pixels.

into five categories and 12 classes. The green mark in Fig.2 shows the proportion of pixels of each class in the total pixels, where pedestrians account for the least, only 0.21%, and roads account for up to 30.78%.

B. Cyclecity

The dataset is acquired by converting the style of Cityscapes to take advantage of its accurate labels. Therefore, the format of the two datasets is exactly the same. Cityscapes is a large urban scene dataset consisting of 5K high-quality annotated images and 20K images with rough annotations. The 5K images with fine annotations are divided into training set, validation set and test set, which include 2975 images, 500 images and 1525 images respectively. Each image has a resolution of 1024×2048 . A total of 30 classes are included, of which 19 classes are used for training and validation. The pixel distribution is shown by the orange mark in Fig.2. The main difference between Cyclecity and Synthesizecarla is that the former dataset refines the vehicles into buses, trucks, tricycles, bicycles, and motorcycles, where the proportion of motorcycle is the smallest, i.e., only 0.1%, and there is no road line class.

C. ACDC-Night [39]

The ACDC is a large-scale semantic segmentation dataset on adverse visual conditions, which consists of 4K images captured under rain, snow, fog and nighttime conditions. There are only 400 open nighttime images with pixel-level annotations, and the resolution is 1080×1920 .

D. Bdd100k-Night [40]

The Bdd100k dataset is known for its large scale and diversity and comprises 100,000 different video clips, six different types of scenes and weather conditions, and three different time periods. There are 10K images with pixel-level annotations for semantic segmentation, of which only 345 are night images. The image resolution of each image in the dataset is 720×1280 , and the formats of annotations are the same as those of Cyclecity.

IV. METHODOLOGY

The proposed method for training involves three kinds of input datasets: a labeled nighttime simulation dataset D_{sys} , a labeled nighttime transferred dataset D_{cycle} and labeled nighttime real-world datasets D_{real} (D_{bdd} and D_{acdc}), which represent Synthesizecarla, Cyclecity, Bdd 100k-night and ACDC-night respectively. Since D_{cycle} and D_{real} share the same annotations, we conducted joint training for them. As shown in Fig.3, the proposed framework consists of two main components: Light Enhancement Network, and Segmentation Network.

A. Light Enhancement Network

Inspired by the previous work [41], we designed an image enhancement network, which learns the dynamic high-order curves of RGB channels through deep CNNs to make pixel-level adjustment for low-quality images with insufficient information. This can compensate for feature loss caused by exposure defects. Unlike the previous work [41], which only depends on pixel fitting, we also introduced semantic information to further address the potential image distortion. In order to enable the network to adaptively select the target position according to the target characteristics and corresponding exposure level, and explore the information related to each pixel, we used the deformable convolution (DCN) commonly used in the detection to optimize the network. DCN [42] uses a learnable offset to represent feature direction of the target region (such as an underexposed vehicle), so that the receptive field is no longer limited to a fixed geometry.

The enhancement network processes the images from three domains and generates the corresponding images respectively. Images from D_{real} and D_{cycle} are fed together into the network; therefore, the two share weights. The detailed structure of this network is shown in Fig.4.

The network consists of seven dynamic convolutional layers, and each convolutional layer is followed by the Relu activation function, except the last one, which uses the tanh activation function to learn best-fitting curves and iteratively generate the final enhanced results. Similarly, following the previous work [41], we removed the BN layer and down-sampling operations that can damage the pixel relation, so that the resolution remains the same. The iterative formula is as follows:

$$LE_n(x) = LE_{n-1}(x) + A_n(x) \times LE_{n-1}(x) \times (1 - LE_{n-1}(x)), \quad (1)$$

where A is a parameter map, and we set $n = 8$, which means eight dynamic adjustments for each pixel.

B. Segmentation Network

The networks used for performing semantic segmentation are usually composed of encoder-decoder and final semantic head. The network reported in previous work [43] is a typical example of this structure. EdgeNet [44] introduced a class-aware edge loss module and a channel-wise attention mechanism in encoder-decoder to improve the accuracy with

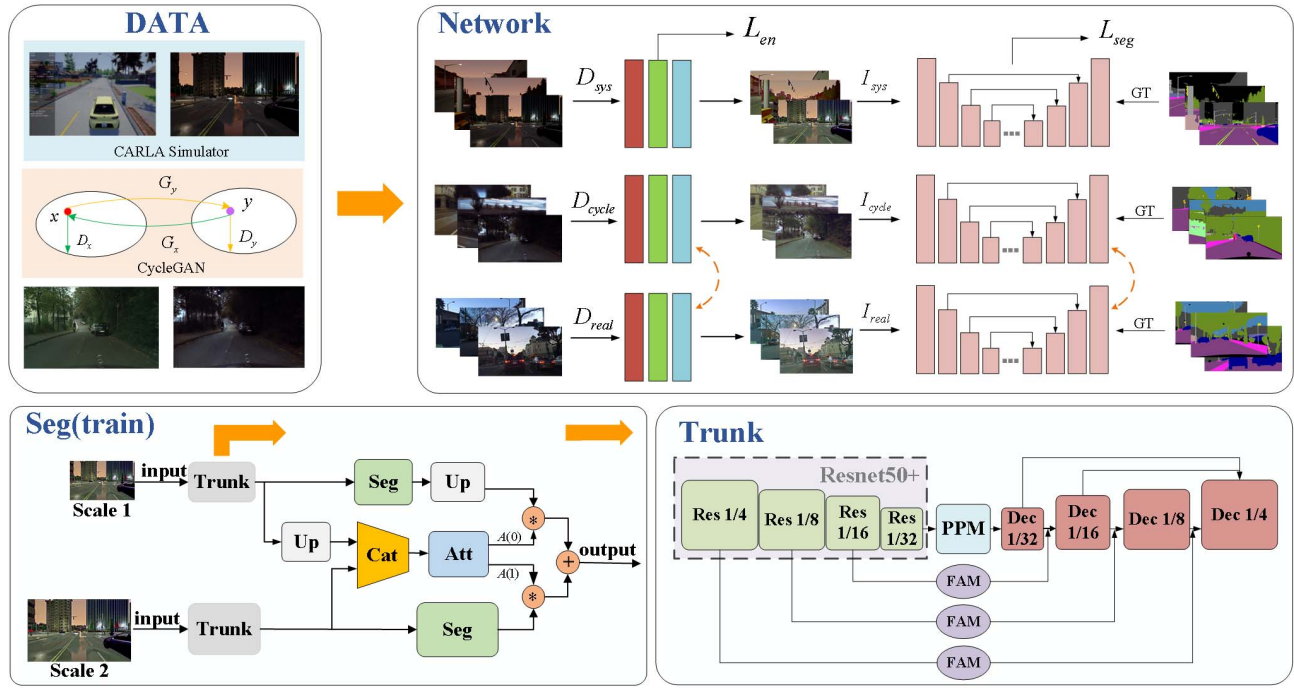


Fig. 3. The first row shows a general overview of the framework, including DATA (three kinds of input datasets D_{sys} , D_{cycle} and D_{real} denote a nighttime simulation dataset, a transferred dataset and nighttime real-world datasets, respectively) and Network (A Light enhancement Network using L_{en} loss and A Segmentation Network using L_{seg} loss). The second row presents the detailed structure of the segmentation network for training. FAM: Feature Alignment Module. PPM: Pyramid Pooling Module.

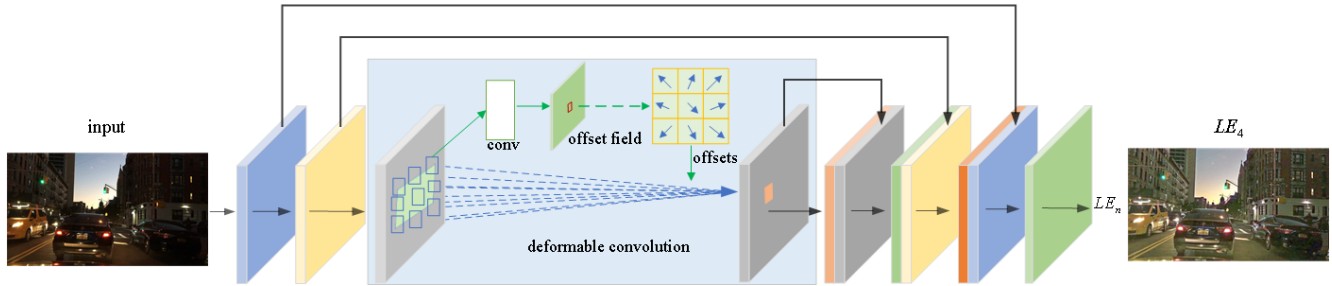


Fig. 4. The structure of the light enhancement network.

no harm to inference speed. Specifically, encoder gradually reduces the image resolution based on the convolution layers, and decoder recovers the details and resolution layer by layer based on deconvolution and comprises different variations of feature pyramid network (FPN) [45]. Besides, decoder also tries to minimize the information loss as much as possible by fusing features at different layers. The semantic header of a segmentation network consists of specialized fully convolutional layers.

At present, SFNet [46] has the optimal results in several daytime segmentation datasets. We modified the detailed structure of SFNet and replaced the backbone with resnet50-CBAM. CBAM [47] is a lightweight module that performs feature-attention in both channel-wise and spatial-wise views. In view of the boundary blur caused by the low semantic contrast of low illumination images, the decoder structure was also redesigned. Considering the effect of image resolution on the segmentation results, we also introduced a multi-scale attention module and further optimized it.

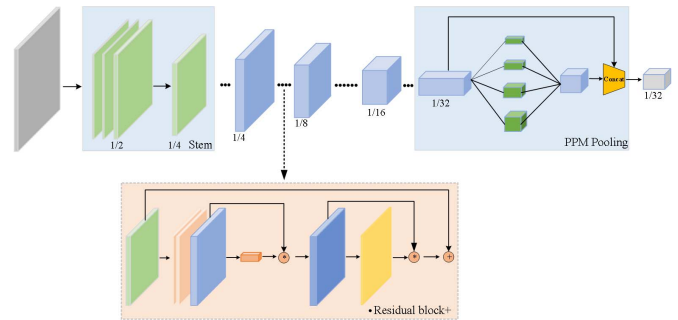


Fig. 5. The architecture of the proposed Encoder.

1) **Encoder**: The encoder is mainly composed of an optimized backbone that extracts the high-level features of the image through multiple convolutions and a pyramid pooling module (PPM) that integrates global context information. The specific details are shown in Fig.5.

In order to achieve a suitable tradeoff between parameters and performance in terms of feature extraction, we selected

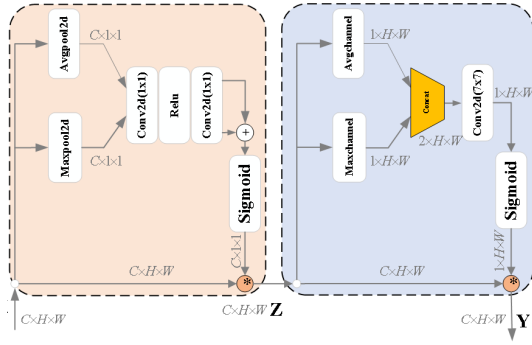


Fig. 6. The details of CBAM module.

ResNet50 as the backbone. We removed the last fully connected layer, and only used the previous 49 convolution layers. In order to further reduce the number of parameters in a convolution and improve the computational efficiency, three small 3×3 convolution kernels are used to replace one large 7×7 convolutional kernel similar to VGG [48]. By stacking multiple small convolution kernels, we can capture more context while keeping the size of the receptive field unchanged. Additionally, an increase in the number of kernels indicates increase in the number of activation functions, non-linearity, and stronger discrimination ability. We also introduced the CBAM to further optimize the residual blocks, as shown in Fig.6. The improved network is called resnet50-CBAM, and improved block+ is shown in Fig.5.

2) *Decoder*: The function of the decoder is to gradually recover the image features using a smaller resolution and stronger semantics obtained by the encoder through successive up-sampling. Generally, the number of times an image is up-sampled is equal to the number of times it is down-sampled. Moreover, a structure similar to the feature pyramid network (FPN) in object detection is also used, and lateral connections are adopted to integrate the high-level strong semantic representation and low-level strong spatial information. However, due to the direct pixel addition between the up-sampled and local features, there are unaligned contexts, which lead to invalid propagation from high-level features to shallow features and misclassifications as well. Feature alignment module (FAM) [46] learns the pixel offset between the feature maps obtained at different adjacent resolutions, which aligns the high-level features with shallow features through local context information, as shown in Fig.7.

3) *Semantic Head*: Finally, we send the high-resolution feature with rich semantic and spatial representations obtained by the decoder to the head consisting of multiple fully convolutional layers. Therefore, the final number of channels in the semantic head is equal to the number of classes.

4) *Multi-Scale Attention*: As the scale of the input image is different, the network may assign pixels to wrong classes leading to false results. The multi-scale inference is typically used to address this problem by using multi-scale images as the input of the same network and obtaining multiple prediction results based on average or maximum pooling during inference. Nevertheless, please note that it is not the

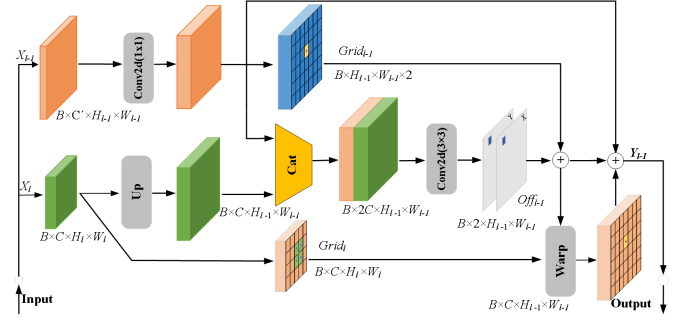


Fig. 7. The details of Feature Alignment Module. X_{l-1} and X_l denote high-resolution of low-level feature and low-resolution of high-level feature maps, respectively. We combined X_{l-1} and transformed X_l to generate field Off_{l-1} , which contains offset in both directions, and warp low-resolution feature map to high-resolution feature map according to bilinear interpolation.

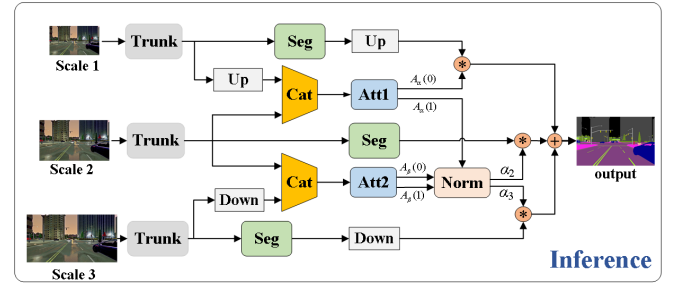


Fig. 8. The multi-scale attention for inference.

best option as it is difficult to decide whether to average multiple predictions or only consider the best scale output.

Inspired by literature [49], we designed hierarchical multi-scale attention. Without being limited to the training scale, we weighed and summed the multi-scale predictions to obtain the best results by learning the soft relative weight between adjacent scales. Different from the hard weighting used by previous work [49] in reasoning, we predicted the mask for each scale of a pair separately, so that the sum of the parameters of predicted attention mechanism channels is the number of scales. This means that the channels of attention for a scale pair is 2, which represents the confidence score of the logit prediction under a single scale, respectively. It is worth noting that since the module adopts hierarchical operations, only one extra scale needs to be trained separately, which is selected as 0.5. The training process is shown in Fig.1. We adopted three scales during inference, namely $0.5\times$, $1\times$, and $2\times$, as shown in Fig.8, which can be expressed mathematically as follows:

$$Attn(\alpha) = F_{1\times 1}(F_{3\times 3}(U_p(X(r = 0.5)) \oplus X(r = 1))), \quad (2)$$

$$Attn(\beta) = F_{1\times 1}(F_{3\times 3}(D_o(X(r = 2)) \oplus X(r = 1))), \quad (3)$$

$$Norm(Z) = \frac{Z}{Attn(\beta)}, \quad (4)$$

$$\begin{aligned} L'''(r=1) &= U_p(L(r=0.5)) \times A_\alpha(0) \\ &\quad + L(r=1) \times A_\beta(0) \times Norm(A_\alpha(1)) \\ &\quad + D_o(L(r=2)) \times A_\beta(1) \times Norm(A_\alpha(1)), \end{aligned} \quad (5)$$

where r denotes the scale factor; $F_{3\times 3}(\cdot)$ and $F_{1\times 1}(\cdot)$ denote the convolution operation using kernels of size 3×3 and 1×1 kernel respectively; $X(\cdot)$ denotes the feature map

TABLE I

THE CONFIGURATION OF SFNET-N. STRIDE INDICATES THE REDUCTION RATE OF THE FINAL OUTPUT SIZE RELATIVE TO THE ORIGINAL INPUT

Layers	Operations	Repeat	Channel	Stride
Light Enhance.	Dconv:3×3	4	32	
	Dconv:3×3	2	64	1
	Dconv:3×3	1	64	
Encoder	Conv:3×3	2	64	2
	Conv:3×3	1	128	2
	Maxpool:3×3	1	128	4
	Layer1(Block+)	3	256	4
	Layer2(Block+)	4	512	8
	Layer3(Block+)	6	1024	16
	Layer4(Block+)	3	2048	32
PPM	Conv:1×1	2	256	32
Decoder	FAM	3	256	32
Head	Conv:3×3		256	
	Conv:1×1	1	classes	1
Scale Attention	Conv:3×3	2	256	
	Conv:1×1	1	2	1

before semantic head; $U_p(\cdot)$ and $D_o(\cdot)$ denote up-sampling and down-sampling; $Attn(\alpha)$ and $Attn(\beta)$ denote the attention maps between two scale pairs, while $A_\alpha(\cdot)$ is a dimension of $Attn(\alpha)$; $Norm(z)$ denotes the relative weight of Z with respect to $Attn(\beta)$; $L(\cdot)$ denotes the logit probability before softmax activation function.

Table I shows the overall configuration of proposed SFNET-N, including operations, repeat, channels and stride, which indicates type of layers and their interconnections. The stride indicates the reduction rate of the final output size relative to the original input.

C. Object Function

In order to enable the network to be effectively segmented, we conducted a series of loss functions, mainly including light loss and segmentation loss.

Light Loss: The light loss is proposed to make the illumination distribution of nighttime images with visual hazards uniform and consistent with the actual scene, including the total variation loss L_{tv} , color constancy loss L_{col} , exposure control loss L_{ex} and spatial consistency loss L_{sp} .

The total variation loss L_{tv} is widely used in image processing [50], [51] to make the image smoother. In this work, we applied the loss function from the RGB channels of the image to preserve the tonal relationship between adjacent pixels of the enhanced images, and the formula is defined as follows:

$$L_{tv} = \frac{1}{N} \sum_{i=1}^N \sum_{c \in (r, g, b)} (|\nabla_x A_n^c| + |\nabla_y A_n^c|), \quad (6)$$

where A_n^c represents learned parameter map of channel c after n iterations; ∇_x and ∇_y denote gradient operations in horizontal and vertical directions respectively.

In order to correct the potential color deviation in enhanced images, the color constancy loss L_{col} proposed in literature [41] is applied to control the color level. Differently, we also considered the difference between the enhanced and

original images, while preventing large gradients. Our loss function is defined as:

$$L_{col} = \frac{1}{N} \sum_{\forall (p, q) \in \varepsilon} \left| \frac{Y^p}{Y^q} - \frac{X^p}{X^q} \right|, \varepsilon = (r, g), (g, b), (b, r), \quad (7)$$

where Y and X denote enhanced images and original images respectively.

To suppress potential underexposure or overexposure in the image, exposure control loss L_{ex} is applied to measure the distance between average intensity of the local regions and well-exposed level E :

$$L_{ex} = \frac{1}{N} \sum_{i=1}^N |\varphi(Y)_i - E|, \quad (8)$$

where φ is 16×16 average pooling operation, E is average intensity value of the enhanced images, which is set to 0.6.

As per the literature report [41], we applied spatial consistency loss L_{sp} to retain the difference between adjacent regions of enhanced and original images, that is, the loss is achieved by calculating the average intensity of local regions of the gray space through fixed convolutions. The formula is defined as follows:

$$L_{sp} = \frac{1}{N} \sum_{i=1}^N \sum_{j \in \Omega} \left(\left| \varphi(Y)_i - \varphi(Y)_i^j \right| - \left| \varphi(X)_i - \varphi(X)_i^j \right| \right)^2, \quad (9)$$

where φ denotes 4×4 average pooling operation, and denotes pixel neighborhoods.

Ultimately, combined with all four loss functions, our illumination loss is defined by:

$$L_{en} = \alpha_{tv} L_{tv} + \alpha_{col} L_{col} + L_{ex} + L_{sp}, \quad (10)$$

where α_{tv} and α_{col} are the weights of loss, set to 20 and 0.5 respectively.

Segmentation loss: We used the multivariate cross entropy loss function commonly used in semantic segmentation as the loss function, which is expressed as follows:

$$L_{seg} = - \sum p(x) \times \log q(x), \quad (11)$$

where, $p(x)$ and $q(x)$ represent ground truth and corresponding prediction, respectively.

Hence, the total loss is computed using parameter and set to 0.1 in Equation (12).

$$L_{total} = \lambda L_{en} + L_{seg}. \quad (12)$$

V. EXPERIMENTS AND RESULTS

A. Authoritative Datasets

In addition to the three kinds of datasets mentioned above for training, we also evaluated nighttime authoritative datasets with a few low-light images of pixel-level annotations, which are described as follows:

TABLE II

THE PER-CATEGORY IOU COMPARISON OF DARK ZURICH-TEST DATASET. EXCEPT FOR EGNET (WHICH IS A FULLY SUPERVISED LEARNING METHOD), THE OTHERS ARE ADAPTATION METHODS. THE BEST RESULTS ARE BLACKENED AND THE SUBOPTIMUM RESULTS ARE UNDERLINED

Methods	road	sidewa.	build.	wall	fence	pole	light.	sign	veget.	terrain	sky	person	rider	car	truck	bus	train	moto.	bicycle	mIoU(%)
AdaptSegNet	86.1	44.2	55.1	22.2	4.8	21.1	5.6	16.7	37.2	8.4	1.2	35.9	26.7	68.2	45.1	0.0	50.1	33.9	15.6	30.4
GCMA	81.7	46.9	58.8	22.0	20.0	41.2	<u>40.5</u>	<u>41.6</u>	64.8	31.0	32.1	53.5	47.5	<u>75.5</u>	39.2	0.0	49.6	30.7	21.0	42.0
MGCDA	80.3	49.3	66.2	7.8	11.0	41.4	38.9	39.0	64.1	18.0	55.8	52.1	<u>53.5</u>	74.7	66.0	0.0	37.5	29.1	22.7	42.5
CDAda	<u>90.5</u>	<u>60.6</u>	67.9	<u>37.0</u>	19.3	<u>42.9</u>	36.4	35.3	<u>66.9</u>	24.4	79.8	45.4	42.9	70.8	<u>51.7</u>	0.0	29.7	27.7	<u>26.2</u>	45.0
DANNNet(PSP)	90.4	60.1	<u>71.0</u>	33.6	<u>22.9</u>	30.6	34.3	33.7	<u>70.5</u>	<u>31.8</u>	<u>80.2</u>	45.7	41.6	67.4	16.8	0.0	<u>73.0</u>	31.6	22.9	45.2
EGNet	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	45.4
Ours	94.3	74.0	79.4	43.8	31.9	43.8	57.9	50.1	73.4	36.1	85.5	60.6	53.6	86.9	8.2	41.2	82.2	45.2	33.7	56.9

1) *Dark Zurich [9]*: This dataset contains 201 annotated images with 19 classes, of which 50 are used for validation and 101 for testing. The resolution of all low-light images is 1920×1080 . Note that Dark Zurich-test is an online test benchmark whose ground truth is not public. In this experiment, we submitted the proposed SFNet-N to an online evaluation website for performance analysis.

2) *Nighttime Driving-Test [10]*: This dataset contains 50 nighttime images with fine annotations, and the resolution is 1920×1080 . The format of annotations is completely consistent with the Cityscapes dataset, with a total of 19 classes.

B. Experiment Details

The semantic segmentation model performs a large number of complex matrix operations and floating-point operations during training process to find the optimal solution. The low-light segmentation model proposed in this work uses Pytorch deep learning framework for distributed data parallel training and synchronized batch normalization based on 64-bit NVIDIA and Ubuntu18.04. We used stochastic gradient descent (SGD) as the optimizer. The momentum and weight decay rates were set to 0.9 and 10-4, respectively. To improve the segmentation performance, the models were pre-trained using ImgeNet [52]. We also used data enhancement strategies during training, including random flip, random rotation, and random crop. The inference was performed at multiple scales, i.e., $0.5\times$, $1\times$, and $2\times$.

The choice of learning rate, batch size, and loss functions can affect the training speed and final segmentation accuracy of the model. A higher learning rate causes a sharp increase in loss, while improving the training speed. Therefore, it is more suitable during the initial stages of training. If the learning rate is set too small, the convergence speed of the model slows down, and it is easier to find the optimal solution of the model. However, the model may overfit. Hence, this learning rate is more suitable during the later stages of training. Considering the aforementioned features, we used a poly adaptive learning rate, i.e., the learning rate decreases continuously with an increase in the number of iterations. This is expressed as follows:

$$l_r = In_l_r \times \left(1 - \frac{epoch}{\max_epoch}\right)^{power}, \quad (13)$$

where, In_l_r denotes the initial learning rate, which is set to 0.1; $epoch$ denotes the number of current epochs; $\max_epoch$ denotes the number of max epochs, which is set

to 300; and $power$ denotes the exponential decay, which is set to 1. Similarly, if the batch size is too large, the training speed is significantly improved, but the utilization of GPU memory is affected, resulting in lower accuracy. Due to hardware limitations, we set the batch size to 4.

C. Evaluation Metrics

We used mIoU to evaluate the performance of the proposed method. In general, the higher the value of mIoU, the better. It is expressed as follows:

$$mIoU = \frac{1}{N} \sum_{n=1}^N \left(\frac{1}{c} \sum_{i=1}^c \frac{p_{nii}}{\sum_{j=1}^c p_{nij} + \sum_{j=1}^c p_{nji} - p_{nii}} \right), \quad (14)$$

where c denotes the total number of classes; N denotes the number of images in the validation set; p_{nii} denotes the number of the pixels which belong to class both in prediction and ground truth in the n^{th} sample; and p_{nij} denotes the number of the pixels which belong to class i while being predicted as class j in the n^{th} sample.

D. Performance Comparison With State-of-the-Art Methods

In this section, we compared the proposed SFNET-N model with the most advanced methods using the two authoritative datasets of Dark Zurich-test and Nighttime Driving-test. Due to the incomplete consistency between the Synthesiscarla and datasets to be compared on classes, we only publicly compared the models trained on transferred dataset D_{cycle} and labeled nighttime real-world datasets D_{real} (D_{bdd} and D_{acdc}).

1) *Comparison on Dark Zurich-Test*: First, we submitted our method online on one open challenge Dark Zurich-test and compared the obtained results with the existing state-of-the-art methods on nighttime segmentation, mainly including CDAda [30], DANNNet [34], GCMA [9], and AdaptSegNet [53], MGCDA [54], EGNet [36]. The previous methods are all domain adaptation methods based on pseudo-labeled or unlabeled images, due to the lack of large-scale datasets on nighttime. However, EGNet [36] adopted the fully supervised learning method, which created a real-scene dataset at night with pixel-level semantic annotations. The comparison results on the per-class IoU performance are reported in Table II, in which the best results are blackened and the suboptimum results are underlined.



Fig. 9. A visual comparison on Dark Zurich-test. The advantages of EGNET are highlighted with white boxes compared with previous methods and the advantages of our method are highlighted with yellow boxes compared with all methods, including EGNET.

Compared with the highest score obtained by existing methods (by EGNET), our method achieved an 11.5% improvement in the test set. Since detailed per-class scores are not provided in EGNET, we also compared our method with the best performance, choosing PSPNet as the backbone among the three DANNet models; an improvement of 11.7% was achieved. Except for the bad effect of truck class, all other classes are improved. To verify the superiority of our method, we selected eight common images in some relevant papers for visualization, as shown in Fig.9. We observed that the proposed method improves the large-scale classes with poor vision quality, such as sky, vegetation, building and sidewalks. This means that our method can alleviate the situation where the distribution of nocturnal features is not obvious to some extent. The edges of small-sized objects also become clearer and smoother, especially in the pose segmentation of pedestrians.

2) *Comparison on Night Driving-Test*: In order to verify the generality of our method, we repeated the above comparison with the night driving-test set (without results in EGNET).

TABLE III
THE COMPARISON OF NIGHT DRIVING-TEST DATASET. THE BEST RESULTS ARE BLACKENED AND THE SUBOPTIMUM RESULTS ARE UNDERLINED

Methods	mIoU(%)
DMAda	36.1
GCMA	45.6
MGCDA	49.4
DANNet(PSPnet)	47.7
CDAda	<u>50.9</u>
Ours	57.4

The results on the performance are reported in Table III. Compared with the best method (CDAda), an improvement of 6.5% was achieved. The visualization results of each model are shown in Fig.10, which were not compared because the CDADA visual results are small. The qualitative results show that our method improves the performance of nighttime semantic segmentation to a certain extent, especially in the recognition of small objects, such as poles, traffic signs, etc.

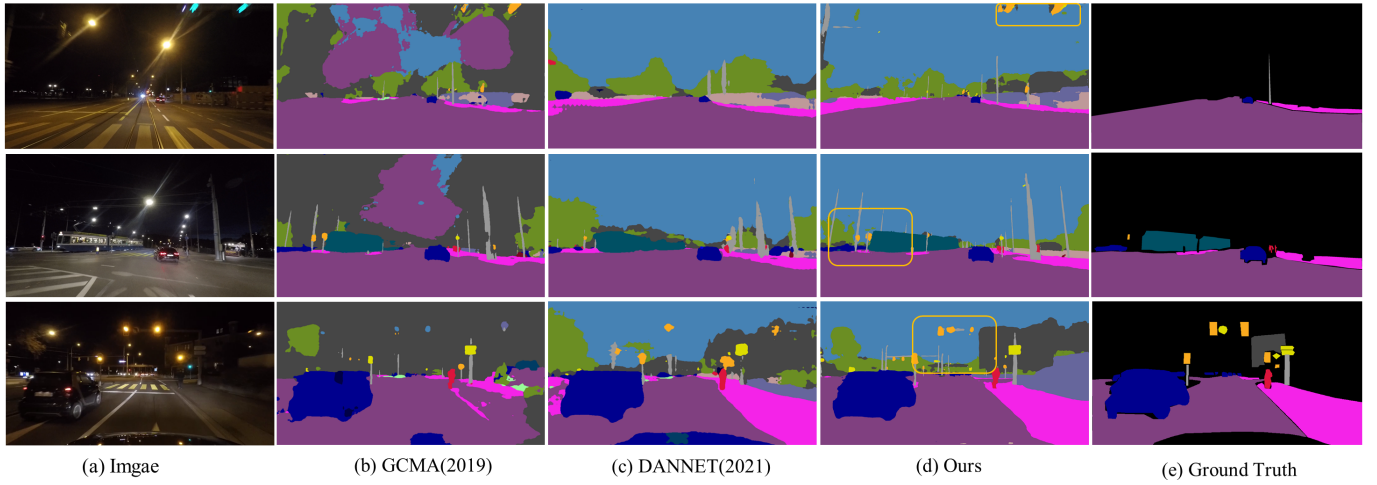


Fig. 10. A visual comparison on Night Driving-test. The advantages of our method are highlighted with yellow boxes.

TABLE IV

THE ABLATION STUDY ON DARK ZURICH-TEST DATASET. SFNET IS CHOSEN AS BASELINE. D_{city} DENOTES CITYSCAPES DATASET IN DAYTIME, OTHERS DENOTE DATASETS IN NIGHTTIME

Methods	D_{city}	D_{cycle}	D_{bdd}	D_{acdc}	mIoU(%)
baseline	✓				25.8
—		✓			34.1
—		✓	✓		40.5
—		✓	✓	✓	49.2
W/En.		✓	✓	✓	52.5
W/DCN		✓	✓	✓	53.9
W/CBAM		✓	✓	✓	55.6
W/Scale A.		✓	✓	✓	56.9

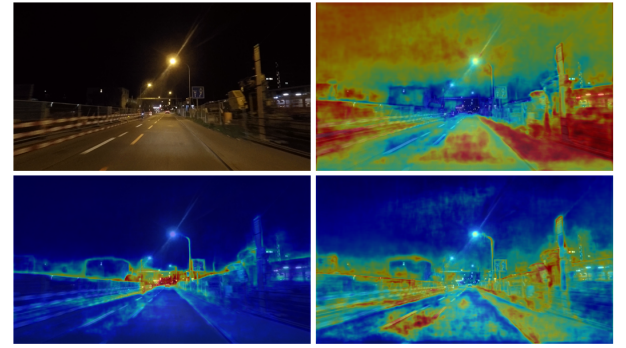


Fig. 12. The visualization of light enhancement network. The visualization of multi-scale attention. In the clockwise direction: actual image, followed by the visualization maps at the scale of 0.5, 1 and 2, respectively.

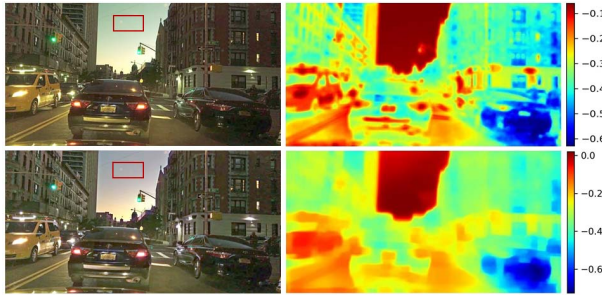


Fig. 11. The visualization of light enhancement network. The first row shows the raw enhanced image and corresponding learned attention map. The second row presents the result with deformable convolution (DCN).

E. Ablation Study

To investigate the effectiveness of each component of the proposed SFNET-N, we tested different model variants trained by combining multiple nighttime datasets on the Dark Zurich-test dataset. The performance results are shown in Table IV. We observed that the daytime baseline model with Resnet-50 as backbone was only 25.8% on the night dataset, which further confirmed our conjecture that the general daytime model cannot be directly adapted to the nighttime domains. Through the simple conversion of daytime datasets, the accuracy was increased by 8.3%. The more nighttime images of joint training, the better the effect of model learning.



Fig. 13. The intelligent vehicle Arrizo-5E for real-vehicle verification.

In addition, the designed illumination enhancement network further improved the performance by increasing the contrast between features, and about 1.4% improvement was achieved by using DCN module. Finally, combined with the attention mechanism and scale attention model, our model could achieve about 56.9% mIoU performance with the dataset.

Fig.11 compares the parameter colormaps learned by the light enhancement network with and without deformable convolution. We observed that dynamic convolution can learn the context information of object areas where the pixel is located

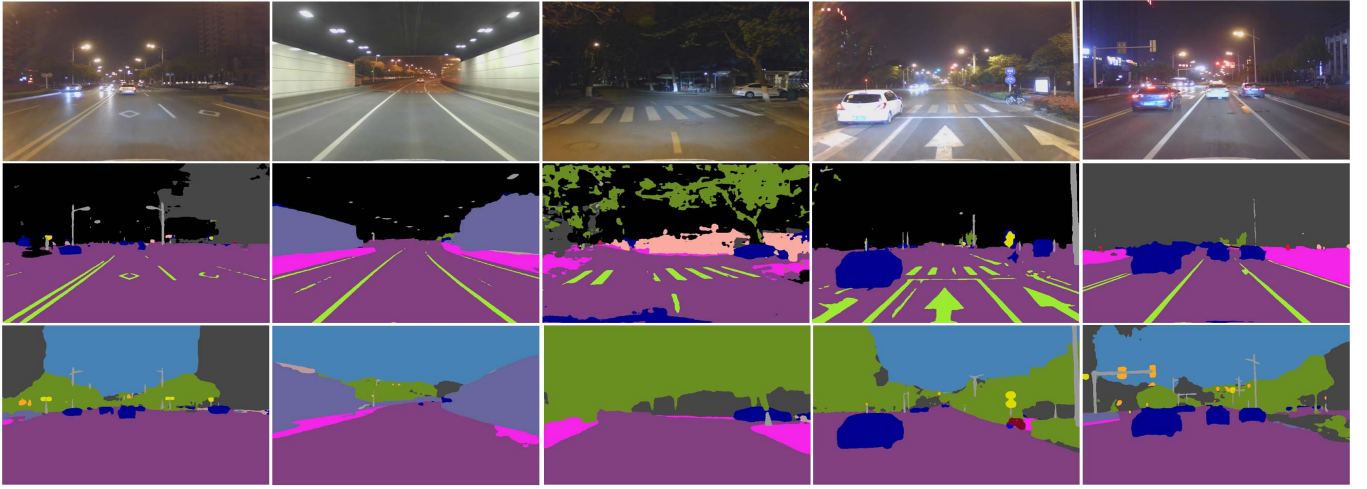


Fig. 14. Comparison of SFNet-N on ZHENJ dataset. From top to bottom: the original image, the result of synthetic data and the prediction of Cyclecity.

TABLE V
THE PERFORMANCE OF PROPOSED SFNET-N. THE INFERENCE
TIME INCLUDES ALL THE TIME OF FEEDING DATA,
LOADING DATA AND SAVING IMAGES

Methods	InputSize	Parameter	Inference/s
SFNet	1920×1080	33.8MB	5
W/En.	1920×1080	34.0MB	4.9
W/DCN	1920×1080	34.1MB	4.9
W/CBAM	1920×1080	34.2MB	4.8
W/Scale A.	1920×1080	39.4MB	3.3

and improve the difference between regions by introducing rough semantic information. Besides, the enhanced images not only improve the contrast, but also generate more realistic scenes, such as the sky. Additionally, we also analyzed the multi-scale attention through examples of visualization of attention maps, as shown in Fig.12. The images are shown in the clockwise direction: actual image, followed by the visualization maps at the scale of 0.5, 1 and 2 respectively. The attention maps learned are consistent with the basic fact that large-scale images pay more attention to the overall features, while the corresponding small-size images are more concerned about fine details. The final experimental data during test are shown in Table V. Compared with the original model, the parameters of SFNet-N are increased by 5.8MB, of which the light enhancement network is only about 0.2 MB. Under the premise that the input sizes are 1920×1080 , our inference speed on single GPU is reduced from 5 Fps to 3.3 Fps. The inference time includes all the time taken for feeding data, loading model and saving images.

F. Quantitative Evaluation on a Real-World Scenario

For semantic segmentation in low-light scenes, this work presents two ideas, i.e., the style transformation of real scene data and virtual data synthesis based on simulation platform. To further improve the night segmentation performance and enhance the driving safety of autonomous vehicles, we designed a network for low illumination scenes and conducted training on two different datasets to obtain the corresponding models. This section mainly describes the

real-vehicle experiments on these two models for the actual scene.

The data were collected from the three scenes, including urban areas, viaducts, and campus of Zhenjiang University by using the Arrizo-5E autonomous vehicle, as shown in Fig.13. The resulting dataset ZHENJ contains 231 images with a resolution of 480×640 . The results of the two models on ZHENJ dataset are shown in Fig.14. From top to bottom, it presents the original image, the result of synthetic data and the prediction of Cyclecity. We observed that the model trained using the synthetic dataset focuses more on the refined details, such as leaf shapes, road line and traffic light, etc., but the segmentation results of vehicles and vegetation are poor. When no clear detailed features are identified, there will be missed detection or even serious false predictions. The model obtained by using the data after style transfer has generally better performance without obvious errors and omissions. Although the traffic signs and vegetation can be recognized in the near view of the image, the edge of segmentation is rough. Additionally, it is evident that the prediction of elevated and urban areas is better compared to that of campus. This may be due to the complex and dynamic campus scenes and lower visibility.

VI. CONCLUSION

In this work, we explored the semantic segmentation of nighttime scenes. To improve the performance, we employed two large-scale low-light datasets with fine annotations for training, namely Synthesis Carla and Cyclecity. Moreover, we proposed an end-to-end framework SFNet-N specially designed for night conditions, which comprises a light enhancement network and a segmentation network. This work first attempted to use deformable convolution to model the illumination enhancement network and optimize the corresponding loss function. As a result, the network improved the authenticity of images after enhancement. At the same time, we used a variety of attention mechanisms in the segmentation network to fully explore the contextual relationship between

pixels and improve the ability of feature extraction as much as possible. Furthermore, experimental results demonstrated that our method outperformed other advanced methods on Dark-Zurich and Night Driving test datasets. Nevertheless, real-vehicle experiments showed that our method still has several shortcomings; especially, there are some errors and omissions in the distant view of images. Also, the real-time performance of the algorithm is not satisfactory. Therefore, we will consider how to enhance the segmentation of small targets at night in the future and improve the inference speed.

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Hai Wang (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees from the School of Instrument Science and Engineering, Southeast University, Nanjing, China.

In 2012, he joined the School of Automotive and Traffic Engineering, Jiangsu University, where he is currently working as a Professor. He has published more than 50 papers in the field of machine vision-based environment sensing for intelligent vehicles. His research interests include computer vision, intelligent transportation systems, and intelligent vehicles.



Yanyan Chen received the B.S. degree from the Nanjing Institute of Technology, Nanjing, China. She is currently pursuing the Ph.D. degree with Jiangsu University.

Her research interests include computer vision, deep learning, and intelligent vehicles.



Yingfeng Cai (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees from the School of Instrument Science and Engineering, Southeast University, Nanjing, China.

In 2013, she joined the Automotive Engineering Research Institute, Jiangsu University, where she is currently working as a Professor. Her research interests include computer vision, intelligent transportation systems, and intelligent automobiles.



Long Chen received the Ph.D. degree in vehicle engineering from Jiangsu University, Zhenjiang, China, in 2002.

His research interests include intelligent automobiles and vehicle control systems.



Yicheng Li received the Ph.D. degree in vehicle engineering from the Wuhan University of Technology, Wuhan, China, in 2018.

He is currently an Assistant Professor at the Automotive Engineering Research Institute, Jiangsu University. His research interests include intelligent vehicle localization, intelligent transportation systems, computer vision, and 3D data processing.



Miguel Angel Sotelo (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Alcalá (UAH), Madrid, Spain, in 2001. He is currently a Full Professor at the Department of Computer Engineering, UAH. His research interests include autonomous vehicles and prediction of intentions. He has served as a Project Evaluator, a Rapporteur, and a Reviewer for the European Commission in the field of ICT for intelligent vehicles and cooperative systems in FP6 and FP7. He is a member of the IEEE ITSS Board of Governors and

Executive Committee. At present, he is the President of the IEEE Intelligent Transportation Systems Society. He has served as the Editor-in-Chief for *IEEE Intelligent Transportation Systems Magazine*, the Editor-in-Chief for the *ITSS Newsletter*, and an Associate Editor for *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS*.



Zhixiong Li (Senior Member, IEEE) received the Ph.D. degree in transportation engineering from the Wuhan University of Technology in 2013. He is currently with the Yonsei Frontier Laboratory, Yonsei University, Seoul, Republic of Korea, and the Faculty of Mechanical Engineering, Opole University of Technology, Poland. He is also the Director of the International Joint Research Centre on Renewable Energy and Sustainable Marine Vehicles. He is the author/coauthor of two books and over 100 papers.

His research interests include dynamic system modeling, renewable energy, and machine learning applications. He is an Associate Editor of *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS* and a Column Editor of *IEEE Intelligent Transportation Systems Magazine*.